Experiment 21: Point Estimation and Confidence Interval

21.Scenario: you are a scientist conducting research on rare elements found in a specific region. Your goal is to estimate the average concentration of a rare element in the region using a random sample of measurements. You will use the NumPy library to perform point estimation and calculate confidence intervals for the population mean.The rare element concentration data is stored in a CSV file named "rare_elements.csv," where each row contains a single measurement of the concentration.

Question:write a Python program that allows the user to input the sample size, confidence level, and desired

level of precision.

Aim: To estimate the population mean and calculate a confidence interval based on user inputs.

Algorithm:

1. Import NumPy and the t-distribution function.
2. Store sample concentration data.
3. Get sample size, confidence level, and desired precision from the user.
4. Calculate the sample mean and confidence interval.
5. Display the mean, confidence interval, and margin of error.

Code:

Experiment 21 – Confidence Interval

```
import numpy as np

from scipy.stats import t

data = np.array([12, 15, 14, 16, 13, 17, 15, 14, 16, 18])

n = int(input("Enter sample size: "))

confidence = float(input("Enter confidence level (e.g. 95): "))

precision = float(input("Enter desired precision: "))

sample = data[:n]
```

```
mean = np.mean(sample)

se = np.std(sample, ddof=1) / np.sqrt(n)

margin = t.ppf((1 + confidence / 100) / 2, n - 1) * se

print("Sample Mean:", round(mean, 2))

print("Confidence Interval:", (round(mean - margin, 2),

                               round(mean + margin, 2)))

print("Margin of Error:", round(margin, 2))

print("Desired Precision Achieved:", margin <= precision)
```

Input:

Sample size: 5

Confidence level: 95

Desired precision: 2

Sample data:

data = [12, 15, 14, 16, 13, 17, 15, 14, 16, 18]

OUTPUT:

Enter sample size: 5

Enter confidence level (e.g. 95): 95

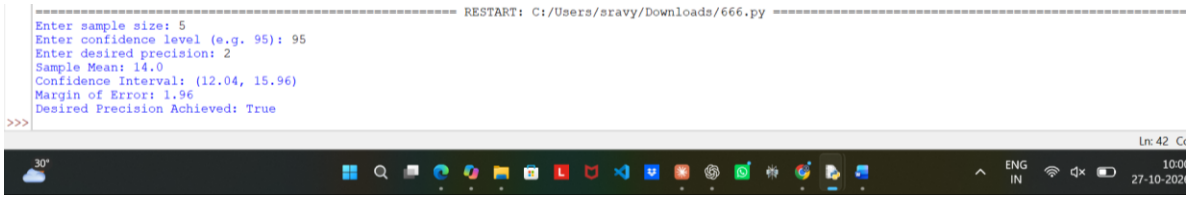
Enter desired precision: 2

Sample Mean: 14.0

Confidence Interval: (12.04, 15.96)

Margin of Error: 1.96

Desired Precision Achieved: True



Result: The sample mean, confidence interval, and margin of error were calculated.

Note: Replace the sample data with measurements from rare_elements.csv for actual results.